

JETLY MINU

jetlyminu@gmail.com | +91 8943206090 | linkedin.com/in/jetlyminu | github.com/Jetlyminu | Kerala, India

Professional Summary

Electronics and Communication Engineer with experience in Python development, embedded systems, and IoT solutions. Skilled in designing and deploying hardware–software integrated systems through academic, freelance, and industry projects. Strong foundation in Python, Embedded C, PCB design, and sensor-based systems, with proven ability to collaborate effectively and deliver reliable, scalable solutions.

Skills Summary

Programming Languages:	Python, C, C++ , Embedded C, VHDL, HTML, CSS
Embedded Systems & IoT:	Arduino, ESP32/ESP8266, STM32, Raspberry Pi, FPGA, Microcontrollers, UART, I2C, SPI, Firmware Development, PCB Design, Sensors, IoT Systems
ML & DL Libraries (Fundamentals):	NumPy, Pandas, Matplotlib, Seaborn, SciPy, OpenCV, Scikit-learn, TensorFlow (Basics), Keras
Machine Learning Concepts:	Natural Language Processing, Convolutional Neural Networks, Recurrent Neural Networks
Tools & Software:	Arduino IDE, MATLAB, Proteus, LTspice, Visual Studio, Blynk

Experience

Freelance Academic & Technical Project Consultant	Feb 2023 – Present
- Designed and implemented college-level projects in Embedded Systems and IoT using Arduino (Uno, Nano, Leonardo, Mega), ESP32, and ESP8266, applying strong problem-solving and analytical skills. - Developed sensor interfacing, motor control, and wireless communication modules, ensuring technically robust and scalable solutions. - Guided 15+ students through project planning, coding, testing, debugging, and documentation, providing mentorship and fostering teamwork. - Mentored Team Follicle Fusion in developing a Smart Hair Comb for MYOSA 3.0 (Singapore), supporting system design, sensor integration, and embedded programming while gaining international collaboration experience.	
Teaching & Workshop Facilitator – Embedded Systems and IoT	Jan 2024 – Present
- Conducted hands-on workshops on Arduino and ESP-based development, explaining complex concepts clearly to diverse student groups. - Demonstrated hardware interfacing, embedded programming, and IoT applications, fostering interactive learning and problem-solving. - Enhanced student understanding via live demonstrations and structured guidance, showcasing leadership and effective communication.	
Chairman – IEEE Computer Society, College of Engineering Attingal	Jan 2024 – Mar 2025
- Led CEAL IEEE activities, coordinating workshops, hands-on sessions, and technical competitions, demonstrating leadership and organizational skills. - Managed student events and knowledge-sharing programs, promoting teamwork, communication, and strategic planning. - Launched student engagement programs, boosting workshop participation by 30%.	

Projects

Wireless Electric Vehicle Charging System (WEvCS) with Bluetooth-Controlled RC Car Mar 2025

- Designed a wireless EV charging system using inductive power transfer for contactless battery charging.
- Developed a Bluetooth-controlled RC car using Arduino Uno R3 and HC-05 for wireless navigation of DC BO motors via L298N/L293D motor driver.
- Integrated wireless power transmitter and receiver modules and programmed functionality in Embedded C using Arduino IDE.

Food Serving Robot using Line Follower Robot (LFR) Method Sep 2024

- Developed an autonomous food-serving robot using Arduino, IR sensors, DC motors, and motor driver modules.
- Implemented line-following and sensor-based navigation algorithms for accurate path tracking and motor control.
- Demonstrated applicability in indoor service automation for hotels, hospitals, and restaurants.

Face Recognition Attendance System using Python and OpenCV Mar 2024

- Developed a software-based attendance system using Python and OpenCV with live camera input.
- Implemented face detection, recognition, and image preprocessing techniques for accurate identification.
- Automated attendance recording to reduce manual errors and prevent proxy attendance.

Education

B.Tech – Electronics and Communication Engineering 2022 – 2025

College of Engineering Attingal, Thiruvananthapuram

B.Tech Minor – Computer Science Engineering 2022 – 2025

College of Engineering Attingal, Thiruvananthapuram

Diploma – Electronics and Communication Engineering 2019 – 2022

MGM Polytechnic College, Kilimanoor, Thiruvananthapuram

Industrial Training

Telecom Technologies — BSNL, Thiruvananthapuram (*5-day training*)

Gained exposure to telecom infrastructure, communication networks, and core telecommunication systems.

Embedded Programming and IoT — Arun Systems (QubyTech), Tirunelveli (*15-day training*)

Hands-on training in embedded programming, IoT system design, sensor interfacing, and microcontroller-based applications.